Seminar Quiz 2

Emprical Analysis for Policymakers

Sciences Po Paris - Reims

30 minutes. No extra papers. You are free to use calculators.

Multiple Choice Questions (MCQs)

1. Which of the following is an assumption of the OLS regression model?

   A) The error term has a mean of zero.

   B) The dependent variable is normally distributed.

   C) The independent variables are perfectly collinear.

   D) The error term is correlated with the independent variables.

2. What does the R-squared value represent in OLS regression?

   A) The proportion of the total variance of the dependent variable explained by the independent variables.

   B) The covariance between the dependent and independent variables.

   C) The correlation between the independent variables.

   D) The sum of squared residuals.

3. In an OLS regression, what does it mean if the residuals are large in magnitude?

   A) The model fits the data well.

   B) The model is overfitting the data.

   C) The independent variables do not explain much of the variation in the dependent variable.

   D) The independent variables explain all the variation in the dependent variable.

4. In an OLS regression, if you add an irrelevant variable, what happens?

   A) The R-squared will remain the same.

   B) The standard error of the regression will decrease.

   C) The R-squared will increase.

   D) The R-squared will decrease.

5. In the context of instrumental variables (IV), which of the following is required for a valid instrument?

   A) It must be correlated with the dependent variable.

   B) It must be correlated with the endogenous regressor but uncorrelated with the error term.

   C) It must be uncorrelated with both the endogenous regressor and the error term.

   D) It must be correlated with the error term.

6. What is the main purpose of using instrumental variables in regression analysis?

   A) To estimate the variance of the error term.

   B) To correct for omitted variable bias in the model.

   C) To account for the interaction between variables.

   D) To ensure that the independent variables are normally distributed.

7. Suppose we use an instrumental variable approach with a weak instrument. What is the consequence of this?

   A) The OLS estimator will be consistent.

   B) The IV estimator will be biased and inconsistent.

   C) The IV estimator will be unbiased and consistent.

   D) The OLS estimator will be efficient.

True/False Questions

1. In OLS regression, the sum of the residuals is always equal to zero.

2. If an instrumental variable is valid, it must be both exogenous and correlated with the endogenous variable.

3. The two-stage least squares (2SLS) method is used to correct for endogeneity in a model.

4. In an OLS regression, the estimated coefficient of an independent variable shows the strength and direction of the relationship between that variable and the dependent variable.

Short Essay Question

You are given the following OLS regression results from a model studying the impact of hours studied (in hours) and participation in an exam preparation course (binary, 1 if participated, 0 otherwise) on final exam scores (out of 100).

Variable	Coefficient	Standard Error
Intercept	50	5
Hours studied	2.5	0.4
Preparation course	8	2

Table 1: Regression results for the effect of study hours and exam prep course on final exam score

Discuss the interpretation of the coefficients in this regression. What is the effect of the preparation course on the final exam score? How would you interpret the coefficient on hours studied? What does the intercept stand for? Comment on the statistical significance of the coefficients (keep your answers brief, use bullet points). (2 points)

Mathematical Questions

1. **OLS Estimation with Binary Independent Variable:** Consider the following simple linear regression model:

$$Y_i = \beta_0 + \beta_1 D_i + u_i$$

where Y_i represents the annual healthcare spending (in dollars) of individual i, and D_i is a binary variable indicating whether the individual has health insurance ($D_i = 1$ if insured, $D_i = 0$ if uninsured).

You are given the following data summary:

Individuals with health insurance ($D = 1$): Mean healthcare spending = \$10,000.

Individuals without health insurance ($D = 0$): Mean healthcare spending = \$6,000.

a) Write down the formulation for $E(Y_i|D_i)$ and calculate β_1. (2 points)

b) We suspect that having health insurance is endogenous to the model, due to omitted variables like "individual health status / risk preferences" and also reverse causality. Show the problem of endogeneity using the formulation you have written in **1.a**. (1 point)

2. **Instrumental Variables:** Continuing from the model above with the assumption that D_i (health insurance) is endogenous, we decide to use an instrumental variable. Consider a hypothetical government policy that randomly selects individuals to receive a substantial health insurance subsidy, which covers a large portion of their insurance premiums. Let Z_i represent whether an individual was randomly selected for the subsidy ($Z_i = 1$ if selected, $Z_i = 0$ if not).

   a) Explain why Z_i (subsidy selection) might be a valid instrument for D_i (health insurance). What conditions must Z_i satisfy to be considered a valid instrument? (2 points)

   b) Write the two equations that would be required to effectively use the instrumental variable in this context. First, write the regression for the estimates of the instrumental variable, then write the original regression using the new estimates (remember to use $\hat{Z}_i$ for estimate values). (2 points)